INTERNAL ENGINEERING BRIEF

Fayyaz Travels Admin

Full technical documentation for the `ftadmin` codebase

Product	Staff admin panel (not the public website)
Codebase	ftadmin — Next.js UI + Fastify API + MySQL `fz`
Production URL	https://fayyaz.travel/panel/
API service	ftbackend (default port 4000)
Document date	14 August 2026
Audience	Engineers joining or maintaining this repository

Fayyaz Travels Admin (`ftadmin`) — Technical Documentation

Product: Internal admin panel for Fayyaz Travels
Codebase: `ftadmin` (Next.js UI + Fastify API, shared MySQL `fz` database)
Public URL (production): `https://fayyaz.travel/panel/`
API service name: `ftbackend`
Document date: 14 August 2026
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This document is a researched inventory of the running system: architecture, folder layout, database, HTTP API, frontend routes, file/function catalog, and every error the backend currently returns. There is **no formal error-ID catalog** in the codebase (no `ERR_*` / `FT-*` codes). Errors are HTTP status + `{ error: string }` payloads. Those messages are listed in §10 Error catalog.

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1. System overview

`ftadmin` is the **staff admin** for Fayyaz Travels. It is not the public website. It manages:

Domain	What staff do
Packages	14-step wizard that writes `add_package` and syncs a stub into the public catalogue `tbl_packages`
Package categories / templates / pricing rules	Taxonomy, reusable skeletons, markup/FX rules
Suppliers	Vendor directory (from the internal vendor sheet)

Domain	What staff do
Blogs / CMS pages	New admin-owned content (`blog_posts`, `cms_pages`) — **not** the legacy `tbl_post` tables
Enquiries	Inbox over the public-site table `tbl_message`
Users / roles	Staff accounts, role matrices, per-user overrides
Settings	Singleton site config, including supplier-quote CC/BCC
Dashboard	Counts and activity mixed from admin tables + legacy website tables
Promotions	UI only (sample data; no API or table yet)

Stack

Layer	Technology
Admin UI	Next.js **16.2.9** App Router, React **19**, Tailwind **4**, shadcn/Radix, TinyMCE, TipTap
API	Fastify **5** on port **4000**, `@fastify/jwt`, `@fastify/multipart`, `@fastify/static`
Database	MySQL / MariaDB, database **`fz`**, `utf8mb4_unicode_ci`, session TZ **`+08:00` (SGT)**
Auth	JWT in `localStorage` (`ft_admin_token`), bcrypt (12 rounds)
Email	Microsoft Graph if configured, else SMTP, else Ethereal (dev)
Jobs	`node-cron` publishes due scheduled blog posts every minute
Validation	Zod (API + forms), Yup is a dependency but Zod is the live path
Process model	Two processes: `tsx server/index.ts` (API) + `next dev` / `next start` (web)

The Next.js app is mounted at **`basePath: /panel`**. `next.config.ts` rewrites `/api/*` and `/uploads/*` to the Fastify process so the browser can call `fetch("/api/...")` without knowing the API port.

---

2. Architecture

```
Browser  ──►  Next.js :3000  (/panel/...)
                 │  rewrites
                 ▼
              Fastify :4000  (/api/..., /uploads/..., /health)
                 │
                 ▼
              MySQL `fz`   (admin tables + legacy tbl_* tables)
                 │
                 └──  uploads/  (images/videos served at /uploads/)
```

Request path in development (`npm run dev`)

1. `scripts/ensure-mysql.mjs` waits for MySQL.
2. `concurrently` starts:
   - `tsx watch server/index.ts` → Fastify on `API_PORT` (default 4000)
   - `next dev` → UI on 3000

3. UI pages call `src/lib/*-api.ts` → `fetch("/api/...")` → Next rewrite → Fastify.
4. Fastify route handlers call `src/lib/*-repository.ts` → `query()` in `src/lib/db.ts`.

Layering (important)

Layer	Location	Runs in
Pages	`src/app/**/page.tsx`	Browser (almost all `"use client"`)
Feature UI	`src/components/<domain>/`	Browser
API clients	`src/lib/*-api.ts`	Browser
Types / Zod	`src/lib/*-types.ts`, `validation.ts`	Shared
Repositories	`src/lib/*-repository.ts`	**API process only** (mysql2)
Routes	`server/routes/*.ts`	API process

There are **no** Next.js `route.ts` handlers. Do not import `*-repository.ts`, `db.ts`, or `email.ts` from React components.

Data ownership

- **Admin-owned tables** are created by `scripts/init-db.sql` and `scripts/migrate-*.sql`.
- **Legacy website tables** (`tbl_packages`, `tbl_message`, `tbl_destinations`, `tbl_testimonials`, `tbl_products`) live in the same `fz` database (or `LEGACY_DB_NAME` if split). This app reads/writes a subset of them. It does **not** own their schema.

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3. Getting started

```
npm install
# copy env: create .env.local with DB_* and JWT_SECRET
npm run db:init          # creates add_package (requires tbl_packages already present)
npm run admin:seed       # bootstrap super_admin (ADMIN_EMAIL / ADMIN_PASSWORD)
npm run dev              # API + Next.js
```

Open `http://localhost:3000` — it redirects to `/panel`.

Useful scripts:

Script	Purpose
`npm run dev`	Ensure MySQL, then API + Next concurrently
`npm run dev:api` / `dev:web`	Start one process
`npm run build` / `start`	Production Next.js
`npm run start:api`	Production Fastify (`tsx server/index.ts`)
`npm run db:init`	Run `scripts/init-db.sql`
`npm run db:migrate:*`	Incremental schema (see §13)
`npm run admin:seed`	Upsert super admin

Script	Purpose
`npm run lint`	ESLint

First user created via **signup** becomes `super_admin`; later signups become `admin`. Prefer `admin:seed` in production.

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4. Environment variables

Loaded by Fastify from `.env.local` then `.env`. Next.js also reads `NEXT_PUBLIC_*`.

Variable	Default	Used by
`DB_HOST`	`localhost`	MySQL pool
`DB_PORT`	`3306`	MySQL pool
`DB_USER`	`root`	MySQL pool
`DB_PASSWORD`	`""`	MySQL pool
`DB_NAME`	`fz`	Admin schema
`LEGACY_DB_NAME`	same as `DB_NAME`	Prefix for `tbl_*` queries if split
`API_PORT`	`4000`	Fastify listen
`API_HOST`	`0.0.0.0`	Fastify listen
`API_LOG_LEVEL`	`info`	Fastify logger
`API_URL`	`http://localhost:4000`	Next.js rewrites (server-side)
`CORS_ORIGIN`	`http://localhost:3000,http://127.0.0.1:3000`	CORS allow-list (comma-split)
`JWT_SECRET`	**required in production** (≥32 chars); hardcoded **dev fallback** otherwise	JWT sign/verify
`JWT_EXPIRES_IN`	`7d`	Token lifetime
`BLOG_PUBLISH_CRON`	`* * * * *`	Scheduled blog publisher
`LEGACY_ASSET_BASE_URL`	`https://fayyaztravels.com/`	Remote image proxy
`LEGACY_ASSET_ROOT_PATH`	unset	Optional local disk root for legacy images
`NEXT_PUBLIC_API_URL`	same-origin `/api`	Browser API base (optional)
`NEXT_PUBLIC_AUDIT_USER`	`"Admin User"`	Fallback audit name
`NEXT_PUBLIC_LEGACY_ASSET_BASE_URL`	`https://fayyaztravels.com`	Client image URLs
`ADMIN_EMAIL` / `ADMIN_PASSWORD`	—	`admin:seed`

Variable	Default	Used by
`GRAPH_CLIENT_ID` / `GRAPH_CLIENT_SECRET` / `GRAPH_TENANT_ID`	—	Microsoft Graph mail
`GRAPH_FROM_EMAIL` / `GRAPH_FROM_NAME`	`notifications@fayyaztravels.com` / `Fayyaz Travels`	Graph from-header
`SMTP_HOST` / `SMTP_PORT` / `SMTP_USER` / `SMTP_PASS` / `SMTP_SECURE`	—	SMTP fallback
`SMTP_USE_ETHEREAL`	on in non-production	Ethereal test inbox
`SKIP_DB_INIT`	—	Docker entrypoint skip
`NODE_ENV`	—	JWT secret enforcement, Ethereal

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5. Folder structure

```
ftadmin/
├── src/                    Next.js admin UI + shared lib
│   ├── app/                App Router pages (basePath /panel)
│   ├── components/         Feature UI + shadcn primitives
│   └── lib/                Types, Zod, API clients, MySQL repositories
├── server/                 Fastify API process
│   ├── index.ts            Bootstrap, CORS, static, route registration
│   ├── plugins/auth.ts     JWT plugin + authenticate decorator
│   ├── lib/                Route helpers (audit, errors)
│   ├── jobs/               Cron (blog publisher)
│   ├── routes/             One plugin per domain
│   └── types/fastify.d.ts  JWT user typing
├── scripts/                SQL migrations, seeds, DB tooling
├── public/                 Logo, TinyMCE (copied on postinstall)
├── uploads/                Runtime media (gitignored); served at /uploads/
├── docker/entrypoint.sh    Container start: optional db:init then CMD
├── Dockerfile
├── docker-compose.yml      Dev: MySQL 8 + app (ports 3000/4000/3306)
├── docker-compose.prod.yml App only, bind 127.0.0.1 for Apache
├── apache-ftpadmin.conf    Proxy /panel, /_next, /api, /uploads
├── next.config.ts          basePath /panel + API rewrites
├── package.json
└── DOCUMENTATION.md        This file
```

`src/app/` — pages

Path	Role
`layout.tsx`	Root layout, Geist fonts, metadata “Fayyaz Travels Admin”
`providers.tsx`	`AuthProvider` → `PermissionsProvider` → `RouteGuard` + Sonner
`globals.css`	Tailwind / theme
`page.tsx`	Dashboard
`login/` `signup/`	Auth (server Suspense wrapper + client form)

Path	Role
`add-package/`	14-step wizard (`layout.tsx` is a Suspense boundary)
`packages/` `packages/[module]/`	Package hub + module views
`blogs/` `blogs/[module]/` `blogs/new/` `blogs/edit/[id]/`	Blog posts + categories
`pages/` `pages/[module]/` `pages/new/` `pages/edit/[id]/`	CMS pages
`suppliers/`	Supplier directory
`promotions/`	Promotions (client sample data)
`enquiry/`	Enquiry inbox
`user-management/`	Staff accounts
`settings/general/`	Site settings
`settings/roles/`	Access-control matrix
`settings/role-management/`	Create/rename/delete roles + employee overrides

There is **no** `src/middleware.ts` and **no** `src/app/**/route.ts`.

`src/components/` — UI by domain

Directory	Role
`layout/`	`AdminShell`, `Sidebar`, `Header`, `BrandLogo`, `ListPageHeader`
`auth/`	Session, route guard, permission context, login card
`dashboard/`	Stat cards, chart, activity, quick actions, content summary
`package/`	Wizard chrome + 14 step forms
`packages/`	List/dashboard (not the wizard)
`blogs/`	Posts + blog categories
`pages/`	CMS pages
`suppliers/`	Vendor directory
`categories/`	Package categories
`templates/`	Package templates
`pricing-rules/`	Markup / FX rules
`enquiries/`	Lead inbox
`users/`	Admin users
`settings/` + `settings/access-control/`	Settings and RBAC UI
`promotions/`	Promotions UI (no backend)

Directory	Role
`ui/`	shadcn/Radix primitives

`server/` — API

File	Role
`index.ts`	Build Fastify, plugins, prefixes, global 503/500 handler, listen
`plugins/auth.ts`	`@fastify/jwt` + `app.authenticate`
`lib/audit.ts`	Audit user from JWT / `x-audit-user`; Super Admin gate
`lib/errors.ts`	`sendRouteError` (used by auth routes)
`jobs/blog-publish-scheduler.ts`	Cron → `publishDueScheduledBlogPosts()`
`types/fastify.d.ts`	JWT payload `{ sub, email, name, role }`
`routes/*.ts`	Domain plugins (see §7)

`scripts/`

Flat folder. SQL files are the schema source of truth. Matching `run-migrate-*.mjs` files execute them. See §13.

`public/`

File	Role
`fayyaz-logo.png`	Brand logo (`/panel/fayyaz-logo.png` because of `basePath`)
`tinymce/`	Copied from `node_modules` by `scripts/copy-tinymce.mjs` (`postinstall`)
`*.svg`	Default Next.js placeholders

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6. Database

Name: `fz`
Charset: `utf8mb4` / `utf8mb4_unicode_ci`
Connection: `src/lib/db.ts` (`mysql2` pool, limit 10, session `time_zone = '+08:00'`)

Audit columns (`created_by`, `updated_by`) store **display names**, not user IDs.

6.1 Tables owned by this app (13)

`add_package` — 14-step package wizard

Created by `scripts/init-db.sql`. One row per wizard package. **1:1** with `tbl_packages` via `package_id`.

Column	Type	Role
`id`	`BIGINT UNSIGNED` AI	PK (this is what `package_category_map.package_id` points at)

Column	Type	Role
`package_id`	`INT` UNIQUE	FK → `tbl_packages.id` **ON DELETE CASCADE**
`status`	ENUM `draft`, `pending_review`, `published`, `archived`, `deleted`	Workflow. Soft delete = `deleted`. Requires `migrate-add-deleted-status.sql`
`current_step`	`TINYINT UNSIGNED` default `1`	Last wizard step (1–14)
`supplier_source` … `review_publish`	`JSON` NULL	One JSON blob per step (NULL = not filled)
`created_by` / `updated_by`	`VARCHAR(255)`	Audit names
`created_at` / `updated_at`	`TIMESTAMP`	

JSON columns: `supplier_source`, `package_basics`, `media_frontend`, `pricing`, `itinerary`, `hotels`, `inclusions`, `booking_policy`, `consultant_view`, `supplier_quote_tool`, `frontend_faqs`, `sales_ads_assets`, `seo`, `review_publish`.

Indexes: status, current_step, created_at, created_by, updated_by.

Create path: insert stub `tbl_packages` (`is_publish = 2`) → insert `add_package`. Updates copy title/slug/SEO back to `tbl_packages`.

`admin_users` — staff login

Column	Role
`id`	PK
`email`	Unique login identity
`password_hash`	bcrypt
`full_name`	Display / JWT `name`
`role`	`VARCHAR(60)` — `admin`, `super_admin`, `owner`, or custom
`is_active`	`1` active, `0` disabled (login fails same as bad password)
`last_login_at`	Set on successful login
timestamps	

`admin_settings` — singleton (`id = 1`)

Site name, company, support email/phone, office address, timezone (default `Asia/Kolkata`), homepage title/description, `maintenance_mode`, `quote_cc_emails` / `quote_bcc_emails` (comma-separated for supplier quotes), `updated_by`.

`role_permissions` — role × module matrix

Column	Role
`role`	Matches `admin_users.role` (logical, no FK)

Column	Role
`module_key`	See §11
`can_view` / `can_edit` / `can_delete`	Flags. Edit/delete imply view when saved

Unique `(role, module_key)`. `super_admin` is treated as full access in code even without rows.

`user_permissions` — per-user overrides

Same flags as role permissions, keyed by `user_id`. **No FK.** `user_id` is `INT UNSIGNED` while `admin_users.id` is `BIGINT UNSIGNED`. Unique `(user_id, module_key)`. Also ensured at runtime by `ensureUserPermissionsTable()`.

`cms_pages` — CMS pages

Title, unique slug, excerpt, HTML `content`, featured image, `page_type` (`about` / `contact` / `legal` / `landing` / …), status `draft|published|archived`, `show_in_nav`, `display_order`, `published_at`, SEO, audit.

`suppliers` — vendor directory

Company, country, city/tour area, operational `status` (`active|inactive|do_not_use`), `relationship_status` (`new`, `under_review`, `with_issues`, `on_probation`, `inactive`, `blacklisted`, `poor`, `average`, `good`, `excellent`), services (CSV), contacts, website, notes, problems, account details, TC POC, relationship owner, sheet date.

`blog_categories` / `blog_posts`

Categories: unique name/slug, description, order, `active|inactive`, SEO, audit.

Posts: title, unique slug, excerpt, HTML, featured image, `category_id` **FK SET NULL**, `author_name`, status `draft|scheduled|published|archived`, `is_featured`, `published_at` (required when scheduled), `tags` JSON, SEO, audit.

Cron: `publishDueScheduledBlogPosts()` sets `scheduled` → `published` when `published_at <= NOW()`.

`package_categories` / `package_category_map`

Categories: name, slug, description, `icon_path`, `image_path`, order, `active|inactive`, `is_featured`, SEO.

Map: composite PK (`package_id`, `category_id`). **`package_id` is `add_package.id`, not `tbl_packages.id`.** Both FKs CASCADE.

`package_templates` — reusable wizard skeletons

Same JSON step columns as `add_package` **except** `review_publish`. Extra: `template_type` (`full|partial`), `included_steps` JSON (step numbers 1–13), `is_default`, `display_order`, `usage_count`. Unique name and slug.

`pricing_rules` — markup / FX / finance thresholds

`scope_type`: `global|destination|currency|supplier_country` + `scope_value`. Currencies, `exchange_rate_type` (`manual|auto|fixed`), `default_exchange_rate`, `rate_lock_days`, `markup_type` (`percentage|fixed|tiered`) + `default_markup_value` / `tiered_markup_json`, `minimum_margin_percent`, `finance_approval_threshold`, `requires_finance_approval`, `is_default` (app clears others), order, audit.

6.2 Legacy tables this app actually queries (5)

These have **no CREATE TABLE in this repo**. Schema is owned by the public website.

`tbl_packages` — public catalogue

PK `id INT`. Wizard create inserts a stub (`p_name`, `p_slug`, empty content, `package_currency = SGD`, `is_publish = 2`, `status = 1`). Later updates sync title, slug, heroes, meta.

Status mapping used by the admin UI (`mapTblPackageStatus`):

1. `is_archive = 1` → `archived`
2. else `is_publish = 2` → `draft`
3. else `status = 2` → `pending_review`
4. else → `published`

Hard-deleting a legacy row also deletes the linked `add_package` row (FK CASCADE). Archiving sets `is_archive = 1`.

Columns the admin cares about: `p_name`, `p_slug`, `feature_img`, `desti_image`, `desti_list`, `day_night`, `package_currency`, `insert_date`, `edit_date`, `is_archive`, `is_publish`, `status`, `user_id`, `meta_title`, `meta_keywords`, `meta_description`.

`tbl_message` — enquiries (there is no `enquiries` table)

This **is** the Enquiry module. `status`: `1` (or NULL) = open, `2` = closed. `message_type` is a string, not a SQL ENUM:

Value	Meaning
`contactus`	Contact Us
popupenquery	Popup / newsletter (legacy spelling)
`contactus_visa`	Visa
`contactus_corporate`	Corporate

Dashboard "newsletter subscribers" = `message_type = 'popupenquery'`. `origin` / `destination` are often destination IDs stored as strings; the list query CASTs them and joins `tbl_destinations`.

`tbl_destinations`

`id`, `d_name`, `status` (`2` = hidden). Used for enquiry origin/destination labels and dashboard destination counts.

`tbl_testimonials`

Dashboard counts + recent activity. Hidden when `status = 2`.

`tbl_products`

Dashboard product counts only (`entry_date`). Hidden when `status = 2`.

6.3 Relationships

```
tbl_packages.id  1 ── CASCADE ── 1  add_package.package_id
                                          │
                                          │ CASCADE
                                          ▼
                              package_category_map.package_id  (= add_package.id)
                                          │
                                          │ CASCADE
                                          ▼
                              package_categories.id

blog_categories.id  1 ── SET NULL ── *  blog_posts.category_id

tbl_destinations.id  ── logical CAST ──  tbl_message.origin / destination

admin_users.role     ── string match ──  role_permissions.role
admin_users.id       ── logical, no FK ──  user_permissions.user_id
```

Declared FKs (four only): `fk_add_package_tbl_packages`, `fk_pcm_package`, `fk_pcm_category`, `fk_blog_posts_category`.

6.4 Tables in `fz` that this app does **not** use

Do not confuse these with the admin schema: `packages` (old wizard, replaced by `add_package`), `travel_enquiries` / `leads` / `travel_plans`, `bookings`, `tbl_booking*`, `tbl_post*`, `tbl_user`, `tbl_settings`, `tbl_itinerary`, `tbl_price`, and other `tbl_*` website tables.

6.5 Seed data

Script	What it inserts
`migrate-add-settings.sql`	`admin_settings` row `id=1`
`seed-admin.mjs`	Super admin from `ADMIN_EMAIL` / `ADMIN_PASSWORD`
`seed-pages.sql`	About Us, Contact Us
`seed-blogs.sql`	Travel Tips / Destinations / News categories + sample posts
`seed-package-categories.sql`	13 catalog categories (Domestic, International, Honeymoon, …)
`seed-package-templates.sql`	Four skeletons (International Group Tour is default full)
`seed-pricing-rules.sql`	Global 18% ZAR→SGD (default) + UAE 22% AED→SGD
`seed-suppliers-from-sheet.mjs`	Optional Google Sheet import

Seeds are idempotent (`INSERT IGNORE` / `WHERE NOT EXISTS` / upsert).

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7. HTTP API

Base: Fastify, default `http://localhost:4000`
Health: `GET /health` → `{ ok: true, service: "ftbackend", database }`
Uploads: `GET /uploads/**` — **public** static files from `uploads/`

Auth: `Authorization: Bearer <jwt>` on almost every `/api/*` route
Audit: mutating routes take `{ user }` from JWT `name`, else header `x-audit-user`

Multipart: max **80 MB**, **1** file. Images → `uploads/images/`, videos → `uploads/videos/`. Filename `{timestamp}-{uuid8}{ext}`.

Auth gates used on the API

Gate	Meaning
Public	No JWT
JWT	Valid bearer token
Super Admin	`request.user.role === "super_admin"`
Access manager	`super_admin` **or** `owner`

The **module permission matrix is not enforced on individual API routes**. Any valid JWT can call package/blog/page CRUD. The matrix is for the Next.js UI. Super Admin (and Owner for roles) is the only extra server-side RBAC.

7.1 Route index

Public

Method	Path	Role
GET	`/health`	Liveness
GET	`/uploads/**`	Serve uploaded files
POST	`/api/auth/signup`	Create account; first user → `super_admin`
POST	`/api/auth/login`	`{ token, user }`

Auth (JWT)

Method	Path	Role
GET	`/api/auth/me`	Reload current user by `sub`

Packages `/api/packages` — JWT (delete is **not** Super-Admin-gated)

Method	Path	Role
GET	`/stats`	Combined wizard + legacy counts
GET	`/expiring-rates`	Rates with `validityTo` within 30 days
GET	`/supplier-quotes`	Packages with non-empty `supplier_quote_tool`
GET	`/recently-published`	Recent published (skip placeholders)
GET	`/combined`	Wizard + unlinked legacy
GET	`/all`	`tbl_packages` only

Method	Path	Role
GET	`/`	Wizard list; `?status=`
POST	`/`	Create wizard package
GET	`/:id`	Get wizard package
PATCH	`/:id`	Update; `?source=legacy` only allows archive
POST	`/:id/supplier-quotes/send`	Email quote request (SMTP/Graph); 503 if mail unconfigured
DELETE	`/:id`	Soft-delete wizard or hard-delete legacy (`?source=legacy`)

Categories `/api/categories` — JWT; DELETE Super Admin

CRUD + list (`search`, `status`, `sort`, `order`, `page`, `limit`). Duplicate name → 409.

Users `/api/users` — JWT; writes Super Admin

`GET /stats`, list, get, create, patch, delete. Cannot deactivate/delete self; cannot deactivate/demote last active Super Admin.

Pricing rules `/api/pricing-rules` — JWT; DELETE Super Admin

Stats, default, list, CRUD. Duplicate name → 409.

Package templates `/api/package-templates` — JWT; DELETE Super Admin

Stats, list, CRUD, `POST /:id/apply` (inactive → 400). Increments `usage_count`.

Blogs `/api/blogs` — JWT; DELETE Super Admin

Stats, authors, list, CRUD. Status `draft|scheduled|published|archived`.

Blog categories `/api/blog-categories` — JWT; DELETE Super Admin

CRUD. Duplicate name → 409.

Pages `/api/pages` — JWT; DELETE Super Admin

Stats, list (`page_type`), CRUD.

Dashboard `/api/dashboard` — JWT

`GET /` with `?chartDays=` **7, 30, or 90** (else 7).

Enquiries `/api/enquiries` — JWT (delete not Super-Admin-gated)

Stats, list (`type`, `status` open/closed), get, `PATCH` status, delete. Reads `tbl_message`.

Settings `/api/settings` — JWT

`GET` / `PUT` singleton. Quote CC/BCC validated as email lists.

Media `/api/media` — JWT

`POST /upload` multipart. Allowed: jpeg/png/gif/webp/svg; mp4/webm/ogg/quicktime.

Permissions `/api/permissions` — JWT; role writes = Access manager

Method	Path	Role
GET	`/`	All role matrices
GET	`/me`	Effective matrix for current user
GET	`/users/:userId`	Role + overrides + effective
PUT	`/users/:userId`	Save diffs from role defaults
DELETE	`/users/:userId`	Clear overrides
PUT	`/:role`	Replace role matrix
PATCH	`/:role`	Rename (not `admin` / `super_admin` / `owner`)
DELETE	`/:role`	Delete custom role; users reassigned to `admin`
POST	`/:role/reset`	Restore catalog defaults

Suppliers `/api/suppliers` — JWT; DELETE Super Admin

Stats, countries, list, CRUD.

Legacy assets `/api/legacy-assets` — JWT

`GET /remote?url=` fetch remote image. `GET /*` try local `LEGACY_ASSET_ROOT_PATH` (path-traversal checked) else remote `LEGACY_ASSET_BASE_URL`. Cache-Control `public, max-age=3600`.

Endpoint count: ~86 named routes plus `/uploads/**`.

---

8. Frontend routes

All paths below are App Router paths. In the browser they are prefixed with `/panel`.

Route	Page	Renders
`/`	Dashboard	Stat cards, 7/30/90 chart, recent activity, quick actions
`/login`	Login	`AuthLayout` + `loginSchema`
`/signup`	Signup	Open registration
`/add-package`	Wizard	`?id=` `?step=` `?templateId=`
`/packages`	Package dashboard	Stats, pipeline, list
`/packages/[module]`	Package modules	See slugs below
`/blogs`	Redirect	→ `/blogs/all`
`/blogs/[module]`	Blog hub	`all`, `drafts`, `scheduled`, `published`, `archived`, `categories`
`/blogs/new` `/blogs/edit/[id]`	Blog editor	`BlogForm`
`/pages`	Redirect	→ `/pages/all`

Route	Page	Renders
`/pages/[module]`	CMS hub	`all`, `drafts`, `published`, `archived`
`/pages/new` `/pages/edit/[id]`	Page editor	`PageForm`
`/suppliers`	Suppliers	`SuppliersPanel`
`/promotions`	Promotions	Sample data only
`/enquiry`	Enquiries	`EnquiryManagementPanel`
`/user-management`	Users	Add User only if `role === "super_admin"`
`/settings/general`	General settings	
`/settings/roles`	Access Control	Permission matrix
`/settings/role-management`	Role Management	Roles + employee overrides

`/packages/[module]` slugs: `all`, `drafts`, `pending-review`, `expiring-rates`, `supplier-quotes`, `categories`, `pricing-rules`, `templates`, `guide`, `archived`. Slug `dashboard` redirects to `/packages`.

---

9. File and function catalog

9.1 Server

File	Exports	Purpose
`server/index.ts`	*(none; `start()` invoked)*	Bootstrap. Internal: `buildServer`, `start`
`server/plugins/auth.ts`	`registerAuthPlugin`	JWT + `app.authenticate`
`server/lib/audit.ts`	`getRequestAudit`, `requireSuperAdmin`	Audit user; 403 unless Super Admin
`server/lib/errors.ts`	`sendRouteError`	DB → 503, else 500
`server/jobs/blog-publish-scheduler.ts`	`startBlogPublishScheduler`	Cron publisher
`server/routes/auth.ts`	`registerAuthRoutes`	signup / login / me
`server/routes/packages.ts`	`registerPackageRoutes`	Package CRUD, quotes, email
`server/routes/categories.ts`	`registerCategoryRoutes`	Package category CRUD
`server/routes/users.ts`	`registerUserRoutes`	Staff CRUD
`server/routes/pricing-rules.ts`	`registerPricingRulesRoutes`	Pricing rule CRUD
`server/routes/package-templates.ts`	`registerPackageTemplateRoutes`	Templates + apply
`server/routes/blogs.ts`	`registerBlogRoutes`	Blog posts
`server/routes/blog-categories.ts`	`registerBlogCategoryRoutes`	Blog categories

File	Exports	Purpose
`server/routes/pages.ts`	`registerPageRoutes`	CMS pages
`server/routes/dashboard.ts`	`registerDashboardRoutes`	Overview
`server/routes/enquiries.ts`	`registerEnquiryRoutes`	`tbl_message` inbox
`server/routes/settings.ts`	`registerSettingsRoutes`	Singleton settings
`server/routes/media.ts`	`registerMediaRoutes`	Multipart upload
`server/routes/permissions.ts`	`registerPermissionRoutes`	Roles + user overrides
`server/routes/suppliers.ts`	`registerSupplierRoutes`	Supplier CRUD
`server/routes/legacy-assets.ts`	`registerLegacyAssetRoutes`	Image proxy

Internal helpers in routes (not exported): `parseStatus`, `parseRole`, `parseScope`, `parseType`, `parseSort`, `parseChartDays`, `parseRelationship`, `parseTemplateType`, `canManageAccess`, `requireAccessManager`, `extensionFor`, `readLocalAsset`, `fetchRemoteAsset`, etc.

9.2 `src/lib` — shared / server repositories / browser clients

Infrastructure

File	Key exports
`db.ts`	`getDbConfig`, `getPool`, `isDbConnectionError`, `verifyDbConnection`, `query`, `DB_TIME_ZONE`
`utils.ts`	`cn` (clsx + tailwind-merge)
`slug.ts`	`slugify`, `uniqueSlug`
`format-date.ts`	`SGT_TIME_ZONE`, `nowMysqlDateTimeSgt`, `toValidDate`, `formatPackageDate`, `formatPackageDateTime`, `formatPackageDateTimeSgt`, `formatPackageTime`, `toDatetimeLocalValue`, `fromDatetimeLocalValue`
`parse-api-response.ts`	`parseApiResponse<T>` — JSON + friendly 502/HTML errors
`audit-context.ts`	`AUDIT_USER_HEADER`, `DEFAULT_AUDIT_USER`, `resolveAuditUser`, `auditContextFromHeader`
`constants.ts`	`NAV_ITEMS`, `STEPS`, `GUIDE_ITEMS`
`types.ts`	`NavItem`, `Step`, `SupplierFormData`
`email-list.ts`	`parseEmailList`, `isValidEmail`, `formatEmailList`, `validateEmailList`
`email.ts`	`isEmailConfigured`, `buildEmailHtml`, `sendMail` (**server only**)
`validation.ts`	Zod schemas for wizard step 1–2, categories, pricing rules, suppliers, templates, blogs, blog categories, pages

Auth

File	Key exports
`auth-config.ts`	`BCRYPT_ROUNDS`, storage keys, `getJwtSecret`, `getJwtExpiresIn`
`auth-types.ts`	`AdminUser`, `AuthTokenPayload`, `SignupInput`, `LoginInput`, `AuthResponse`
`auth-storage.ts`	`getStoredToken`, `setStoredAuth`, `getStoredUser`, `clearStoredAuth`, `getUserInitials`
`auth-validation.ts`	`signupSchema`, `loginSchema`, `createUserSchema`, `updateUserSchema`
`auth-api.ts`	`signupRequest`, `loginRequest`, `fetchCurrentUser`, `getAuthHeadersForApi`
`auth-repository.ts`	`countAdminUsers`, `getAdminUserStats`, `listAdminUsers`, `findAdminUserById`, `deleteAdminUser`, `updateAdminUser`, `createAdminUserByAdmin`, `findAdminByEmail`, `findAdminById`, `createAdminUser`, `verifyAdminPassword`, `touchAdminLastLogin`, `toAuthTokenPayload`

Permissions / users

File	Key exports
`permission-types.ts`	`ACCESS_MODULES`, `ACCESS_MODULE_KEYS`, `NAV_HREF_MODULE`, `PACKAGE_STEP_PERMISSION_KEYS`, `defaultPermissionForRole`, `buildDefaultMatrix`, `hasModulePermission`
`permission-api.ts`	`getPermissions`, `getMyPermissions`, `getUserPermissions`, `updateUserPermissions`, `resetUserPermissions`, `updateRolePermissions`, `resetRolePermissions`, `renameRole`, `deleteRole`
`permission-repository.ts`	`ensureRolePermissionsSeeded`, `getPermissionsPayload`, `getRolePermissionMatrix`, `getUserPermissionOverrides`, `setUserPermissionOverrides`, `clearUserPermissionOverrides`, `getEffectivePermissionMatrix`, `diffFromRoleDefaults`, `updateRolePermissions`, `resetRolePermissions`, `renameRole`, `deleteRole`
`role-catalog.ts`	`normalizeRoleKey`, `isValidRoleKey`, `getKnownRolesFromStorage`, `rememberKnownRoles`, `rememberKnownRole`, `setKnownRoles`, `forgetKnownRole`, `replaceKnownRole`
`user-types.ts`	Filters, `ROLE_LABELS`, `formatRoleLabel`
`user-api.ts`	`getUserStats`, `listUsers`, `createUser`, `updateUser`, `deleteUser`

Packages

File	Key exports
`package-types.ts`	All wizard JSON types, `PackageRecord`, `PACKAGE_STEP_FIELDS`, status helpers
`package-modules.ts`	`PACKAGE_MODULES`, `PACKAGE_MODULE_SLUGS`
`package-api.ts`	`createPackage`, `updatePackage`, `getPackage`, `listPackages`, `listAllPackages`, `getPackageStats`, `listExpiringPackages`, `listSupplierQuoteRequests`, `deletePackage`, `archivePackage`, `duplicatePackage`, `sendSupplierQuoteEmail`, `getPackageEditHref`, `isWizardPackage`, `getPackageRecordKey`
`package-repository.ts`	`listTblPackages`, `listPackages`, `getPackageById`, `createPackage`, `updatePackage`, `deletePackage`, `deleteLegacyPackage`, `archiveLegacyPackage`, `listAllPackages`, `getPackageStats`, `listExpiringPackages`, `listSupplierQuoteRequests`, `listRecentlyPublishedPackages`
`package-images.ts`	`normalizeLegacyImagePath`, `getPackageImagePath`, `resolvePackageImageUrl`, `resolvePackageProxyUrl`, `getPackageImageUrl`, `getPackageImageSources`
`package-detail-sections.ts`	`PACKAGE_DETAIL_SECTIONS`, `getFieldLabel`

Templates / categories / pricing

File	Key exports
`package-template-api.ts`	`getPackageTemplateStats`, `listPackageTemplates`, `getPackageTemplate`, `createPackageTemplate`, `updatePackageTemplate`, `deletePackageTemplate`, `applyPackageTemplate`, `listActiveTemplates`
`package-template-repository.ts`	Same operations + `resolveTemplateSlug`, `buildApplyTemplatePayload`, `listActiveTemplateOptions`
`category-api.ts`	`listCategories`, `getCategory`, `createCategory`, `updateCategory`, `deleteCategory`, `listActiveCategories`
`category-repository.ts`	Same + `resolveCategorySlug`, `syncPackageCategories`, `getPackageCategoryIds`, `getPackageCategoryNames`, `listActiveCategoryOptions`
`pricing-rules-api.ts`	`listPricingRules`, `getPricingRulesStats`, `getDefaultPricingRule`, `getPricingRule`, CRUD
`pricing-rules-repository.ts`	Matching SQL CRUD

Content (blogs, pages)

File	Key exports
`blog-api.ts`	`listBlogPosts`, `getBlogStats`, `listBlogAuthors`, `getBlogPost`, CRUD

File	Key exports
`blog-repository.ts`	Same + `resolveBlogPostSlug`, `publishDueScheduledBlogPosts`
`blog-modules.ts`	`BLOG_MODULES`
`blog-category-api.ts` / `blog-category-repository.ts`	Blog category CRUD + slug resolve
`page-api.ts`	`listPages`, `getPageStats`, `getPage`, CRUD
`page-repository.ts`	Same + `resolvePageSlug`
`page-modules.ts`	`PAGE_MODULES`

Suppliers / enquiries / dashboard / settings / media / promotions

File	Key exports
`supplier-api.ts`	`listSuppliers`, `getSuppliersStats`, `listSupplierCountries`, CRUD
`supplier-repository.ts`	Matching SQL
`supplier-types.ts`	Status/relationship enums, service helpers, `buildSupplierSummary`
`supplier-excel.ts`	`suppliersToExcelCsv`, `downloadSuppliersExcel`
`supplier-quote-email.ts`	`buildQuoteEmailSubject`, `buildQuoteEmailBody`
`enquiry-api.ts`	`getEnquiryStats`, `listEnquiries`, `getEnquiry`, `updateEnquiryStatus`, `deleteEnquiry`
`enquiry-repository.ts`	SQL on `tbl_message`
`enquiry-types.ts`	`enquiryDisplayName`, `enquiryTypeLabel`, `isEnquiryOpen`
`dashboard-api.ts`	`getDashboardOverview`
`dashboard-repository.ts`	`getDashboardOverview` (server)
`dashboard-data.ts`	`STAT_CARD_CONFIG`, `QUICK_ACTIONS`, `buildStatCards`, `formatRelativeTime`, `mapActivities`
`settings-api.ts`	`getSettings`, `updateSettings`
`settings-repository.ts`	`getAdminSettings`, `updateAdminSettings`
`media-api.ts`	`uploadMediaFile`
`promotion-types.ts`	`SAMPLE_PROMOTIONS`, `packageOptionKey` — **no API**

9.3 Components (feature files)

Layout: `AdminShell`, `Sidebar` (filters nav with `can(module, "view")`), `Header` (search, enquiry notifications, logout), `BrandLogo`, `ListPageHeader`.

Auth: `AuthProvider` / `useAuth`, `PermissionsProvider` / `usePermissions`, `RouteGuard`, `AuthLayout`.

Wizard (`components/package/`): `PageHeader`, `Stepper`, `ProgressCard`, `GuideCard`, `PackageAuditCard` (`formatAuditSummary`), `PackageRichTextEditor`, `DuplicatePackageModal`, plus one form per step: `SupplierForm`, `PackageBasicsForm`, `MediaFrontendForm`, `PricingForm`, `ItineraryForm`, `HotelsForm`, `InclusionsForm`, `BookingPolicyForm`, `ConsultantViewForm`, `SupplierQuoteTool`, `FrontEndFaqsForm`, `SalesAdsAssetsForm`, `SeoForm`, `ReviewPublishForm`. Shared: `StepFormBaseProps`, `PACKAGE_STEP_SAVE_SELECTOR`.

List panels: `*Panel`, `*ListTable`, `*Form`, `*DetailsModal`, `*StatsGrid` for packages, blogs, pages, suppliers, categories, templates, pricing rules, enquiries, users.

Settings access-control: `RolesPanel`, `RoleManagementDesignPanel`, `useRolePermissions`, `useEmployeeOverrides`, `RoleManagementSection`, `RoleFormModal`, `PermissionMatrixSection`, `EmployeeAccessDrawer`, `UserDesignationManagementSection`, `ActivityLogTable` (UI-only log).

UI primitives (`components/ui/`): Alert, AlertDialog, Badge, Button, Card, Checkbox, Dialog, DropdownMenu, Icon, Input, Label, NativeSelect, Popover, RadioGroup, Select, Separator, Sheet, Skeleton, Toaster, Switch, Table, Tabs, Textarea, ToggleSwitch, Tooltip.

9.4 App pages

Each `src/app/**/page.tsx` is a thin client wrapper around a panel, except login/signup (forms) and `add-package/page.tsx` (wizard orchestrator: load, save, step change, template apply, duplicate).

---

10. Error catalog

10.1 There are no application error IDs

The codebase does **not** define codes such as `ERR_AUTH_001` or `FT-404`. Clients must branch on:

1. **HTTP status**
2. JSON body `{ error: string }` (sometimes also `{ message }`, `{ details }` for Zod)
3. Node/MySQL **driver codes** on the server (`ECONNREFUSED`, `PROTOCOL_CONNECTION_LOST`)

`parseApiResponse` maps failed responses to `Error.message = body.error` (or `"Validation failed"` if `details` is present).

10.2 HTTP statuses used by the API

Status	When
200	Success (GET/PATCH/PUT/DELETE)
201	Created (signup, most POSTs)
400	Bad id, bad query, Zod validation, business rule
401	Missing/invalid JWT, bad login
403	Not Super Admin / not access manager
404	Resource not found
409	Duplicate email / name / role

Status	When
500	Uncaught / repository throw
503	MySQL down, or email not configured (quote send)

10.3 Global / infrastructure

Status	Message	Meaning
503	`Database is unavailable. Start MySQL and try again.`	`isDbConnectionError` — `ECONNREFUSED` or `PROTOCOL_CONNECTION_LOST`
500	`Internal server error`	Unhandled non-Error throw
500	*(Error.message)*	Unhandled `Error` (global handler)
401	`Unauthorized`	JWT missing or invalid (`app.authenticate`)
403	`Forbidden` + `Only Super Admin can perform this action`	`requireSuperAdmin`
403	`Forbidden` + `Only Super Admin or Owner can manage roles`	`requireAccessManager` (permissions routes)

Client-side (`parseApiResponse`) when the proxy returns HTML/502:

Thrown message	Meaning
`Invalid JSON response from server`	`Content-Type: json` but body is not JSON
`API server is unavailable. Make sure the backend is running on port 4000.`	502/503 or plain `Internal Server Error`
`API route not found (N). Restart the dev server if you recently added new API routes.`	HTML 404 from Next instead of Fastify
`Unexpected non-JSON response from server`	Success-looking non-JSON
`Validation failed`	`{ details }` without `error`
`Request failed (N)`	Fallback

Auth React hooks:

Thrown message	Meaning
`useAuth must be used within AuthProvider`	Hook used outside provider
`usePermissions must be used within PermissionsProvider`	Same

10.4 Auth (`/api/auth`)

Status	Message	Meaning
400	Zod field errors in `details`	Invalid signup/login body
409	`An account with this email already exists`	Signup email taken
401	`Invalid email or password`	Bad password, unknown email, **or inactive user** (same message on purpose)
401	`Unauthorized`	`/me` with bad/missing user

Repository: `Failed to create admin user` → 500.

10.5 Users (`/api/users`)

Status	Message	Meaning
400	`Invalid user ID`	Non-numeric `:id`
400	`You cannot deactivate your own account`	Self disable
400	`You cannot change your own role`	Self role patch
400	`You cannot delete your own account`	Self delete
400	`Cannot delete the last active Super Admin`	Last Super Admin protection
400	(Zod / last-Super-Admin demote text)	Role/active guards
404	`User not found`	
409	`An account with this email already exists`	Create duplicate

10.6 Generic resource IDs (pattern repeated across domains)

Replace `X` with the resource:

Status	Message
400	`Invalid package id` / `Invalid blog post ID` / `Invalid category ID` / `Invalid page ID` / `Invalid template ID` / `Invalid pricing rule ID` / `Invalid supplier ID` / `Invalid enquiry ID` / `Invalid user id`
400	`Invalid status filter` (packages)
404	`Package not found` / `Blog post not found` / `Category not found` / `Page not found` / `Template not found` / `Pricing rule not found` / `Supplier not found` / `Enquiry not found`

10.7 Packages extra

Status	Message	Meaning
400	`Unsupported update for legacy package`	PATCH legacy with anything other than archive

Status	Message	Meaning
400	`At least one recipient is required`	Quote email with empty `to`
400	Zod `details`	Quote payload invalid
503	Email-not-configured message	`isEmailConfigured()` false

10.8 Enquiries extra

Status	Message
400	`Status must be open or closed`

10.9 Media

Status	Message	Meaning
400	`No file uploaded`	Empty multipart
400	(unsupported type text)	MIME not in image/video allow-list
400	`Upload aborted or file too large`	> 80 MB or aborted stream

10.10 Legacy assets

Status	Message
400	`Image url is required`
400	`Asset path is required`
404	`Asset not found`

10.11 Permissions extra

Status	Message	Meaning
400	`Invalid role`	Role slug failed Zod
400	`This role cannot be renamed` / `cannot be deleted`	Protected: `admin`, `super_admin`, `owner`
400	Unknown module key in body	Not in `ACCESS_MODULE_KEYS`
409	`A role with that name already exists`	Rename collision

10.12 Settings

Status	Message
400	Zod / `validateEmailList` text for quote CC/BCC

10.13 Repository throws (surface as HTTP 500 unless mapped to 409)

These `throw new Error(...)` strings become `{ error: "<message>" }` on 500 (or 409 when the route catches duplicates):

Message	Source
`Unable to generate a valid slug`	blog / blog-category / page / category / template slugify
`Schedule date/time (SGT) is required for scheduled posts`	`blog-repository`
`Blog title is required`	`blog-repository`
`Failed to load blog post after create`	`blog-repository`
`Category name is required`	category + blog-category repos
`A category with this name already exists`	same (often mapped 409)
`Failed to load category after create`	same
`Page title is required`	`page-repository`
`Failed to load page after create`	`page-repository`
`Template name is required`	`package-template-repository`
`A template with this name already exists`	same
`Failed to load template after create`	same
`Only active templates can be applied`	apply endpoint → 400
`Rule name is required`	`pricing-rules-repository`
`A pricing rule with this name already exists`	same
`Failed to load pricing rule after create`	same
`Company name is required` / `Country is required`	`supplier-repository`
`Failed to load created supplier`	`supplier-repository`
`Failed to load package after create`	`package-repository`
`Cannot delete the last active Super Admin`	`auth-repository`
`Failed to create admin user`	`auth-repository`
`JWT_SECRET must be set ...`	`getJwtSecret()` in production

Message	Source
Graph/SMTP send failures	`email.ts` (`Failed to get Graph access token`, `Graph sendMail failed`, `Microsoft Graph is not configured`)

10.14 Node / MySQL driver codes (server-side only)

Code	Meaning in this app
`ECONNREFUSED`	MySQL not running → treated as 503
`PROTOCOL_CONNECTION_LOST`	Dropped MySQL connection → 503
Duplicate key `ER_DUP_ENTRY`	Usually wrapped as 409 “already exists” on email/name, otherwise 500

---

11. Authentication and permissions

Login flow

```
POST /api/auth/login
  → findAdminByEmail → bcrypt compare + is_active
  → touchAdminLastLogin → jwtSign({ sub, email, name, role })
  → localStorage ft_admin_token / ft_admin_user
```

- No refresh token. No logout/revoke endpoint (client clears storage).
- JWT role is **not** re-read from DB on each request except `GET /api/auth/me`. Role changes apply to the API after re-login (UI may refresh via `/me`).

Frontend tree

```
RootLayout
  └ AppProviders
      ├ AuthProvider          localStorage + /api/auth/me
      │   └ PermissionsProvider   GET /api/permissions/me
      │       ├ RouteGuard       login required (not module ACL)
      │       └ Toaster
```

RouteGuard public paths: `/login`, `/signup` only. Authenticated users hitting those go to `/`. Unauthenticated users on private paths go to `/login?next=`.

`can(moduleKey, action)`: Super Admin with an empty matrix fail-opens to `true`. Otherwise `hasModulePermission`. Sidebar uses `NAV_HREF_MODULE`. **Promotions is not mapped**, so it always shows. Wizard steps use `PACKAGE_STEP_PERMISSION_KEYS`.

Permission module keys

Panel: `dashboard`, `blogs`, `blog_categories`, `pages`, `packages`, `package_categories`, `suppliers`, `pricing_rules`, `package_templates`, `media`, `enquiries`, `settings`, `users`, `roles`.

Package builder: `supplier_source`, `package_basics`, `media_frontend`, `pricing`, `itinerary`, `hotels`, `inclusions`, `exclusions`, `booking_policy`, `consultant_view`, `supplier_quote_tool`, `frontend_faqs`, `sales_ads_assets`, `seo`,

`review_publish`.

`exclusions` is a permission key only — there is no `add_package.exclusions` column (exclusions live inside `inclusions` JSON).

Default matrix: `super_admin` all true. Other roles: view+edit on most modules, **no** view/edit/delete on `users` and `roles`, **no** delete anywhere else until granted.

API enforcement vs UI: deletes of categories, pricing rules, templates, blogs, blog categories, pages, suppliers, and all user writes require Super Admin. Role CRUD requires Super Admin or Owner. Package delete and enquiry delete are JWT-only.

---

12. Package wizard (14 steps)

Route: `/add-package` (`src/app/add-package/page.tsx`)

Query	Meaning
`?id=`	Existing `add_package.id`
`?step=1-14`	Jump to a form
`?templateId=`	Apply template, `createPackage`, rewrite URL to `?id=`

Persistence: first save → `POST /api/packages` → URL gains `?id=`. Later saves → `PATCH` with the JSON field from `PACKAGE_STEP_FIELDS`. Save Draft clicks `[data-package-step-save]` and forces `status: "draft"`.

Step	Title	Department	Form	JSON column
1	Supplier Source	Product	`SupplierForm`	`supplier_source`
2	Package Basics	Product	`PackageBasicsForm`	`package_basics`
3	Media & Front-End	Product + Marketing	`MediaFrontendForm`	`media_frontend`
4	Pricing	Product + Finance	`PricingForm`	`pricing`
5	Itinerary	Product	`ItineraryForm`	`itinerary`
6	Hotels	Product + Ops	`HotelsForm`	`hotels`
7	Inclusions	Product	`InclusionsForm`	`inclusions`
8	Booking Policy	Ops + Finance	`BookingPolicyForm`	`booking_policy`
9	Consultant View	Sales Lead	`ConsultantViewForm`	`consultant_view`
10	Supplier Quote Tool	Consultants	`SupplierQuoteTool`	`supplier_quote_tool`
11	Front-End FAQs	Product + SEO	`FrontEndFaqsForm`	`frontend_faqs`
12	Sales & Ads Assets	Marketing + Sales	`SalesAdsAssetsForm`	`sales_ads_assets`
13	SEO	SEO Team	`SeoForm`	`seo`
14	Review & Publish	Final Approver	`ReviewPublishForm`	`review_publish`

Step 14 actions: save draft / submit → `pending_review` / publish → `published`. Templates copy steps 1–13 only. Duplicate creates a new draft titled `Copy of …`.

Wizard packages and legacy `tbl_packages` rows can share numeric IDs. The UI keys records as `wizard-{id}` / `legacy-{id}` (`getPackageRecordKey`).

---

13. Scripts, seeds, and migrations

npm script	SQL / JS	Effect
`db:init`	`init-db.sql`	Create `add_package` (needs `tbl_packages`)
`db:migrate:audit`	`migrate-add-audit-fields.sql`	`created_by` / `updated_by` on wizard
`db:migrate:admin`	`migrate-add-admin-users.sql`	`admin_users`
`db:migrate:admin-role-string`	alter role column to `VARCHAR`	Custom roles
`db:migrate:permissions`	`migrate-add-permissions.sql`	`role_permissions`
`db:migrate:permissions-role-string`	alter role column	
`db:migrate:user-permissions`	`migrate-add-user-permissions.sql`	`user_permissions`
`db:migrate:categories`	package categories + map	
`db:migrate:pricing-rules`	`pricing_rules`	
`db:migrate:templates`	`package_templates`	
`db:migrate:settings`	`admin_settings`	
`db:migrate:quote-emails`	CC/BCC columns	
`db:migrate:blogs`	`blog_categories` + `blog_posts`	
`db:migrate:blog-scheduled`	add `scheduled` to post status	
`db:migrate:pages`	`cms_pages`	
`db:migrate:suppliers`	`suppliers`	
`db:seed:suppliers`	sheet import	
`admin:seed`	bootstrap Super Admin	
`test:audit` / `test:categories`	HTTP smoke tests	

Other SQL without a dedicated npm alias (run the matching `run-migrate-*.mjs` or the SQL by hand): `migrate-add-deleted-status.sql`, `migrate-alter-supplier-relationship-status.sql`, `migrate-alter-supplier-active-status.sql`.

Tooling: `ensure-mysql.mjs` (dev wait), `check-db.mjs`, `copy-tinymce.mjs`, `promote-super-admin.mjs`, `migrate-to-shadcn.mjs` (one-off).

14. Deployment

Docker (dev): `docker-compose.yml` — MySQL 8 (`fz`/`fz`) + app, ports 3000/4000/3306, `uploads` volume. Entrypoint runs `init-db.mjs` unless `SKIP_DB_INIT=true`.

Docker (prod): `docker-compose.prod.yml` — app only, `.env.production`, bind `127.0.0.1:3000` and `:4000`.

Apache: `apache-ftpadmin.conf`

Public path	Upstream
`/panel/`	Next.js `:3000/panel/`
`/_next/`	Next.js `:3000/_next/`
`/api/`	Fastify `:4000/api/`
`/uploads/`	Fastify `:4000/uploads/`

The public Fayyaz site stays on `/`. `X-Forwarded-Proto` / `X-Forwarded-Host` are set for the app.

Dockerfile: Node 22 Alpine, `npm ci` + `next build`, runner copies `.next`, `src`, `server`, `scripts`, installs `tsx` + `concurrently`, exposes 3000 + 4000, CMD starts API + web.

---

15. Conventions and known gaps

Conventions

- New tables: `BIGINT UNSIGNED` PKs, `TIMESTAMP`, audit **names**, JSON step blobs.
- Legacy `tbl_*`: `INT` PKs, `DATETIME` `insert_date`/`edit_date`, integer `status` (`2` often means hidden/closed).
- Dates in the admin UI are **Singapore time** (`Asia/Singapore`, `+08:00`), even though `admin_settings.timezone` defaults to `Asia/Kolkata`.
- Browser talks to Fastify via `*-api.ts`. Repositories stay on the API process.
- Soft-delete for wizard packages is `status = 'deleted'` (not a `deleted_at` column).

Known gaps / drift

1. **No error-ID system** — see §10.
2. **Promotions** has UI + `SAMPLE_PROMOTIONS` only; no table, repository, or API.
3. **Module ACL is UI-only** — a JWT can call most write APIs regardless of `can_edit`.
4. **Package / enquiry DELETE** are not Super-Admin-gated.
5. `add_package.status` `deleted` must be applied via `migrate-add-deleted-status.sql`. If that migration is missing, soft-delete writes fail.
6. `user_permissions.user_id` is `INT` vs `admin_users.id` `BIGINT`; no FK.
7. **Signup is public** — first account is Super Admin; lock this down in production if the panel is reachable.
8. **Uploads are public** if the URL is known (`GET /uploads/**` has no JWT).
9. **JWT secret** has a hardcoded development fallback; production must set `JWT_SECRET`.
10. `exclusions` **permission key** does not map to a wizard column.
11. **ActivityLogTable** in role management is local UI state, not an audit table.

12. Legacy website tables (`tbl_post`, `tbl_user`, …) are **not** the admin blogs/users.

Appendix A — npm scripts (from `package.json`)

`dev`, `dev:api`, `dev:web`, `build`, `build:api`, `start`, `start:api`, `lint`, `db:init`, `db:migrate:audit`, `db:migrate:admin`, `db:migrate:admin-role-string`, `db:migrate:permissions-role-string`, `db:migrate:categories`, `db:migrate:pricing-rules`, `db:migrate:templates`, `db:migrate:settings`, `db:migrate:quote-emails`, `db:migrate:permissions`, `db:migrate:user-permissions`, `db:migrate:blogs`, `db:migrate:blog-scheduled`, `db:migrate:pages`, `db:migrate:suppliers`, `db:seed:suppliers`, `admin:seed`, `postinstall`, `test:audit`, `test:categories`.

Appendix B — Wizard JSON ↔ `tbl_packages` sync (mental model)

```
Staff fills wizard
    → add_package JSON columns (source of truth for admin)
    → selected fields copied to tbl_packages (public catalogue stub)
    → package_category_map links add_package.id → package_categories
```

Legacy-only packages (rows in `tbl_packages` with no `add_package`) still appear in `/packages/all` via `listAllPackages` / `listTblPackages`. Editing them in the wizard is not supported; the UI can archive or delete them as legacy.

End of document. When you add a module: table + migrate SQL + repository + route plugin + `*-api.ts` + `ACCESS_MODULES` entry + nav item.